

COMPENDIUM OF THE NATIONAL ENERGY MANAGEMENT OF INDONESIA



**The National Energy Council
2012**

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1. INDONESIA'S PROFILE

A. Geography

Indonesia lies between latitudes 11° S and 6° N, and longitudes 95° E and 141° East. Consists of 17,508 islands, about 6,000 of them inhabited, spread on both sides of the equator. The largest are Java, Sumatra, Borneo (shared with Brunei and Malaysia), Papua (along with Papua New Guinea) and Sulawesi.

B. General Information

Indonesia shares land borders with Malaysia on Borneo, Papua New Guinea on the island of Papua and East Timor on the island of Timor. Indonesia shares maritime boundary in the form of narrow straits with Singapore, Malaysia, and the Philippines to the north, and with Australia to the south. The capital is Jakarta, located on the island of

Java, and is the largest city followed by Surabaya, Bandung, Medan and Semarang. With a land area of 1,919.4 thousand km², making Indonesia the 16th largest country in the world in terms of land area.

C. Population

Indonesia's population density is 134 people per km², which is ranked the 79th most populous in the world, whereas Java is the most populous island in the world, with the population density of 940 inhabitants per km².

Based on the 2010 Population Census, the number of Indonesia's population is 237.6 million people, consisting of 119.5 million man and 118 million woman. Distribution of the population is concentrated in Java Island by 58 percent, followed by Sumatra Island with 21 percent. West Java, East Java and Central Java are the top three provinces with the most

populous, which amounted to 43 million people, 37.5 million people, and 32.4 million people respectively. While the province of North Sumatra Province is the highest number of population outside the island of Java, with a population of 13 million people.

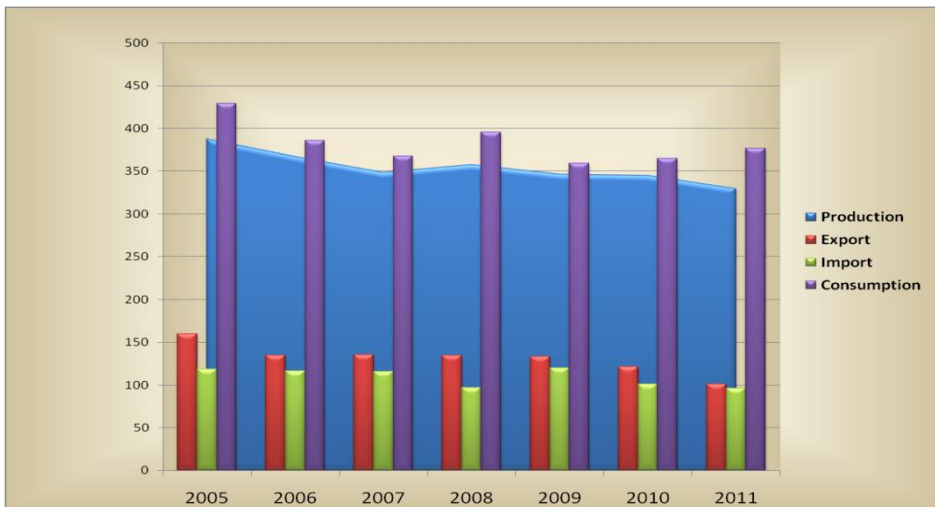
D. Economy

- GDP 2011 at 2000 constant prices: IDR 2.5 trillion.
- GDP/Capita 2011: IDR 29.57 million
- GDP/Capita Growth 2010-2011: 10 percent
- The Indonesian economy in 2011 grew by 6.5 percent compared to 2010.

2. PRIMARY ENERGY SUPPLY BY SOURCES, 2011

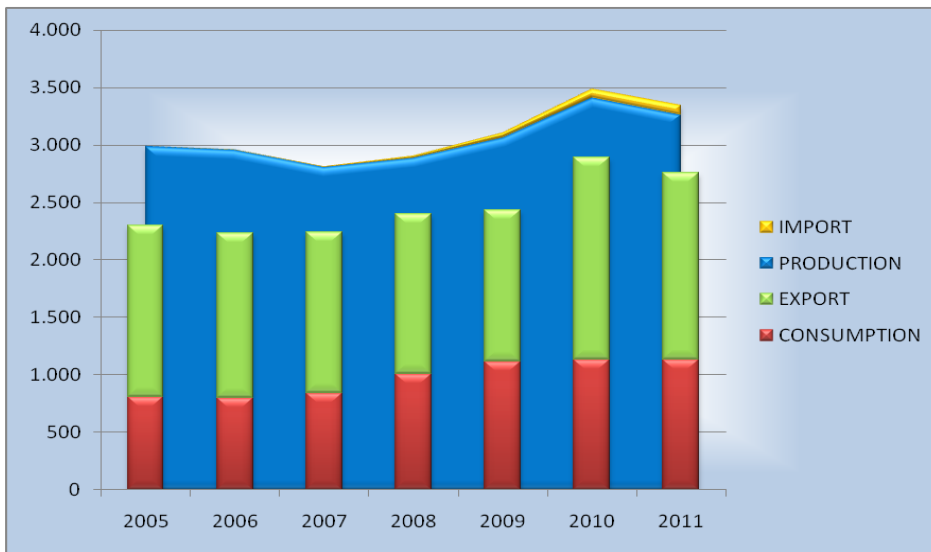
a. Oil

(thousand barrels)



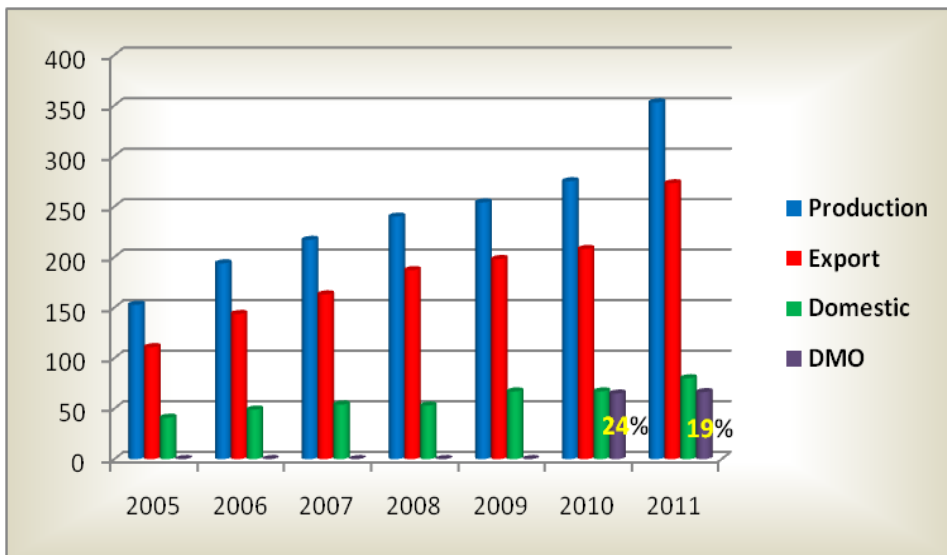
b. Natural Gas, 2011

(thousand MMSCF)



c. Coal,201

(million ton)



3. ENERGY FLOW 2011

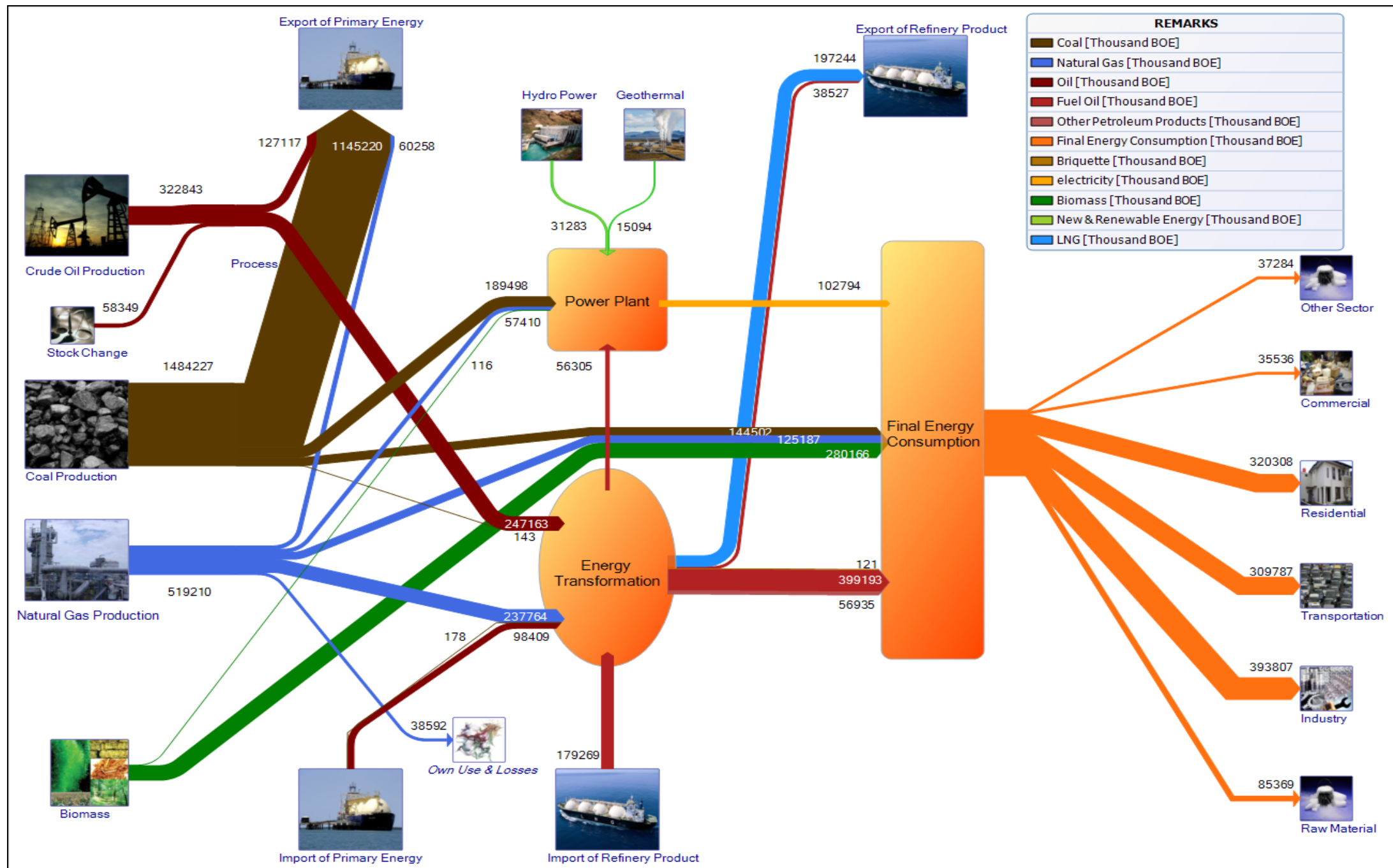
Definition

Energy flow is a flow and the total energy derived from own production or importation, processing, conversion and transformation of energy to meet the energy needs of the various sectors of users up to the level of end-use energy.

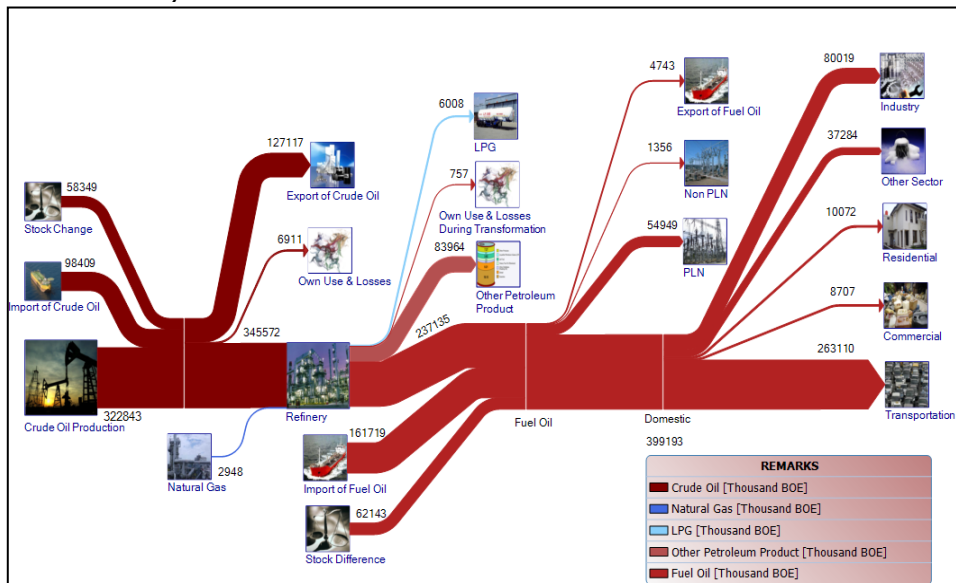
Objective

Cognizant of the size or the total amount of energy produced and the national energy supply and its utilization by the various sectors of users.

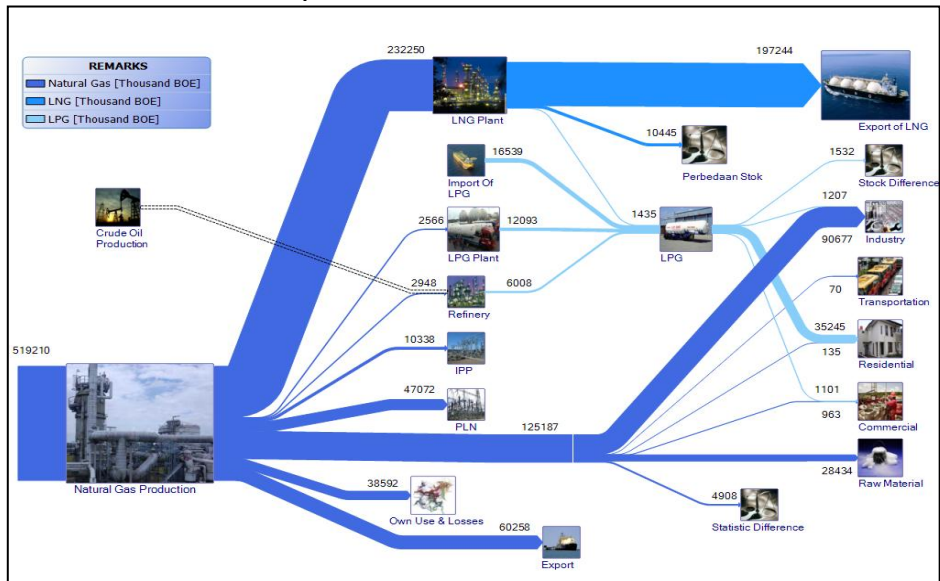
Energy Flow 2011



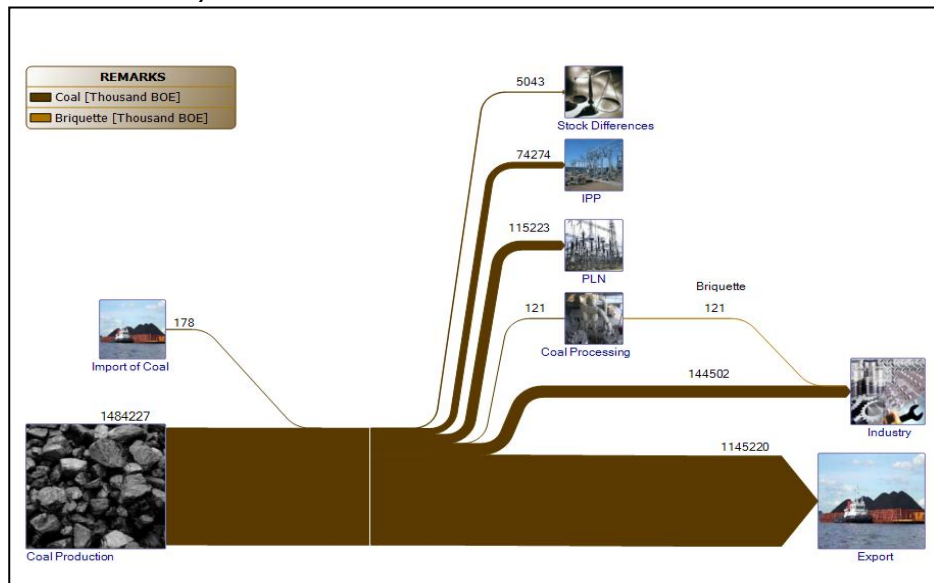
A. Flow of Oil, 2011



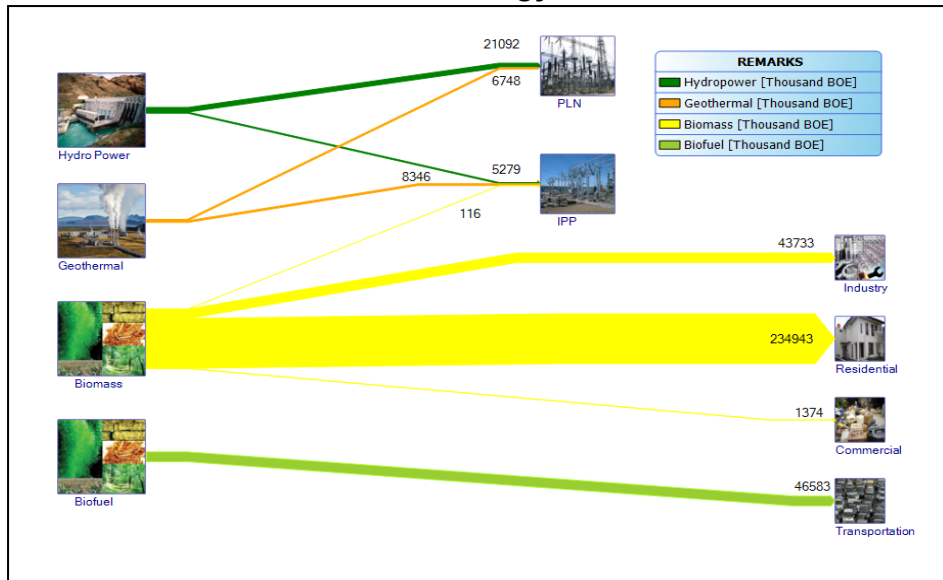
B. Flow of Natural Gas, 2011



C. Flow of Coal, 2011



D. Flow of New and Renewable Energy, 2011

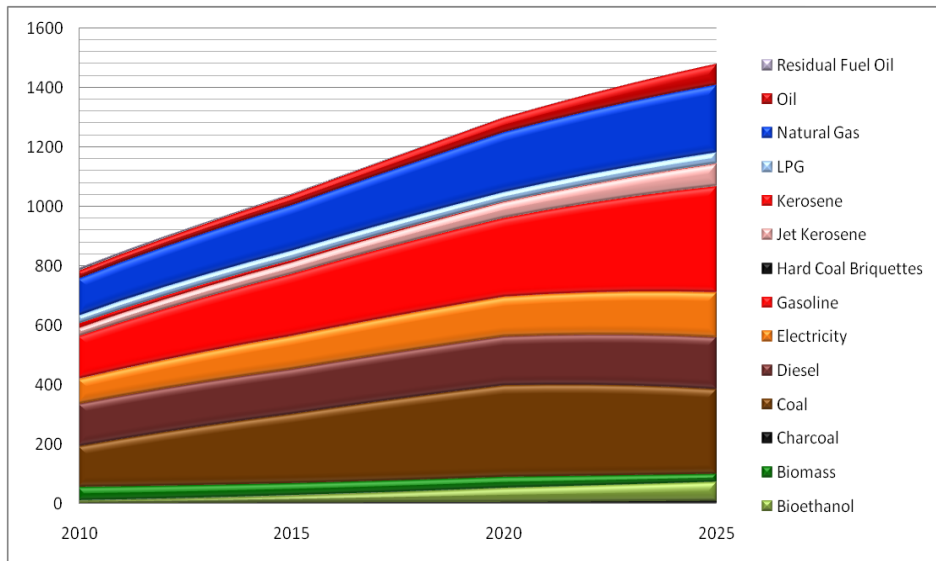


4. NATIONAL ENERGY DEMAND PROJECTION (2010-2025)

Domestic energy demand continues to rise in line with the economic growth and the increase of population. This increase in energy demand, is projected to continue to grow up in the period 2010-2025 based on The Presidential Regulation No.5/2006. Currently, the national energy demand is still dominated by fuel oil and especially diesel and gasoline premium.

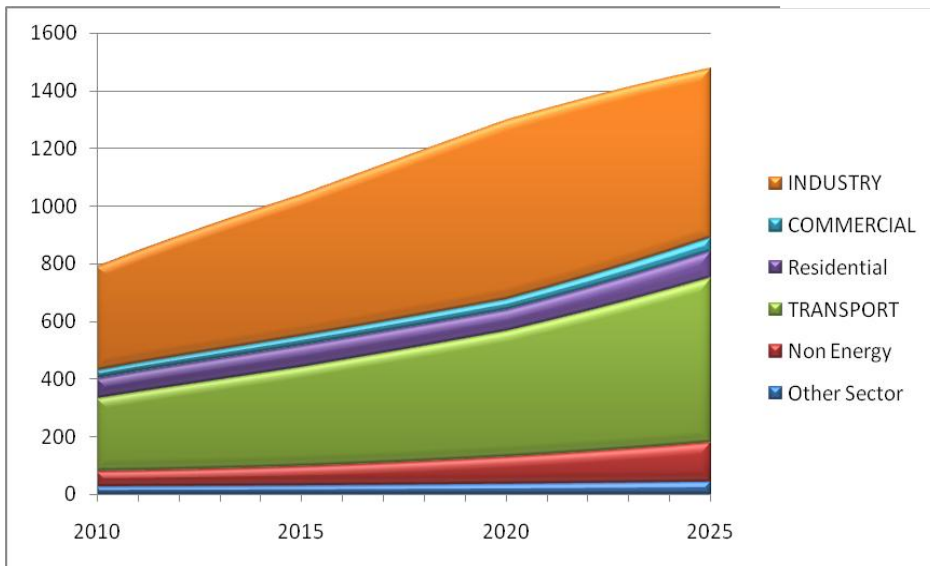
A. Domestic Energy Demand, 2011

(Million BOE)



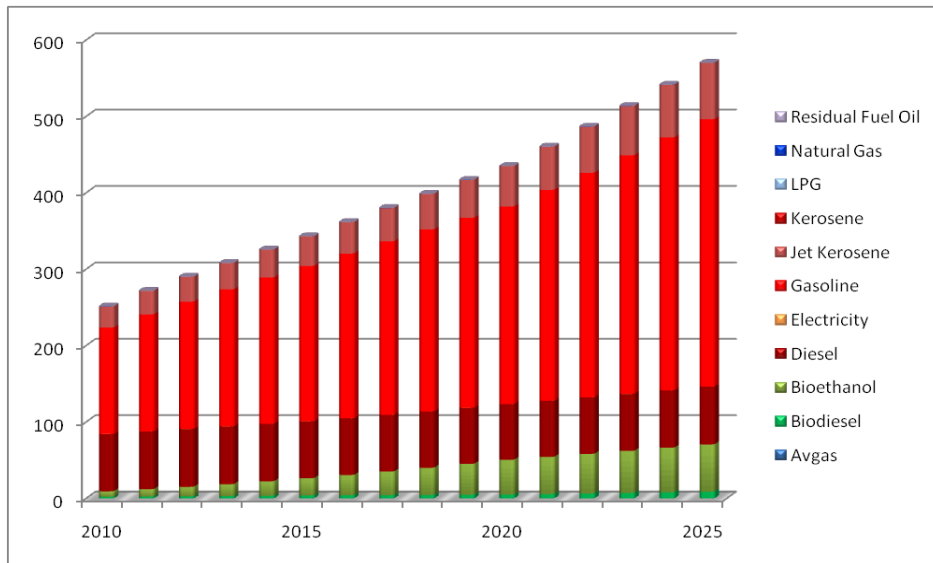
B. Energy Consumption, 2011

(million BOE)



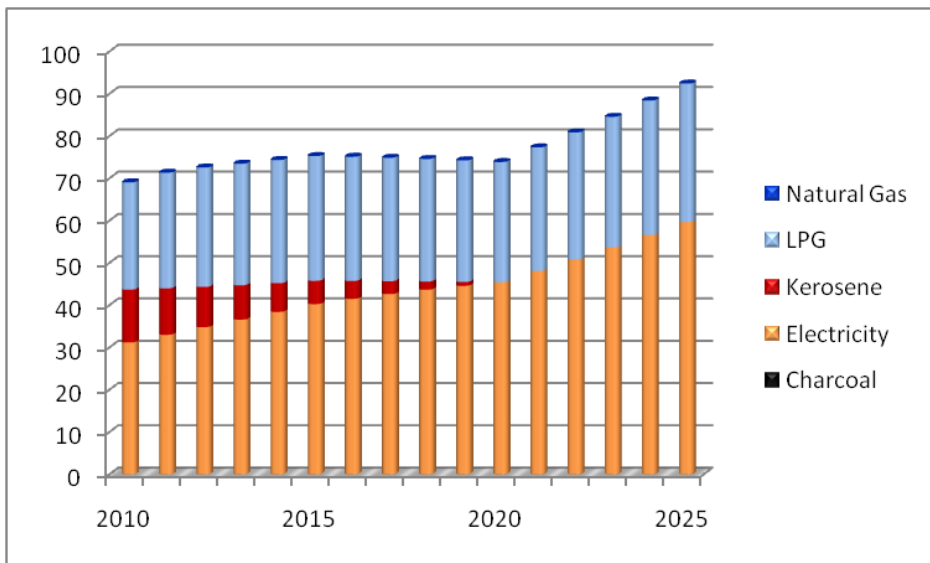
1. Energy Consumption in Transportation Sector, 2011

(million BOE)



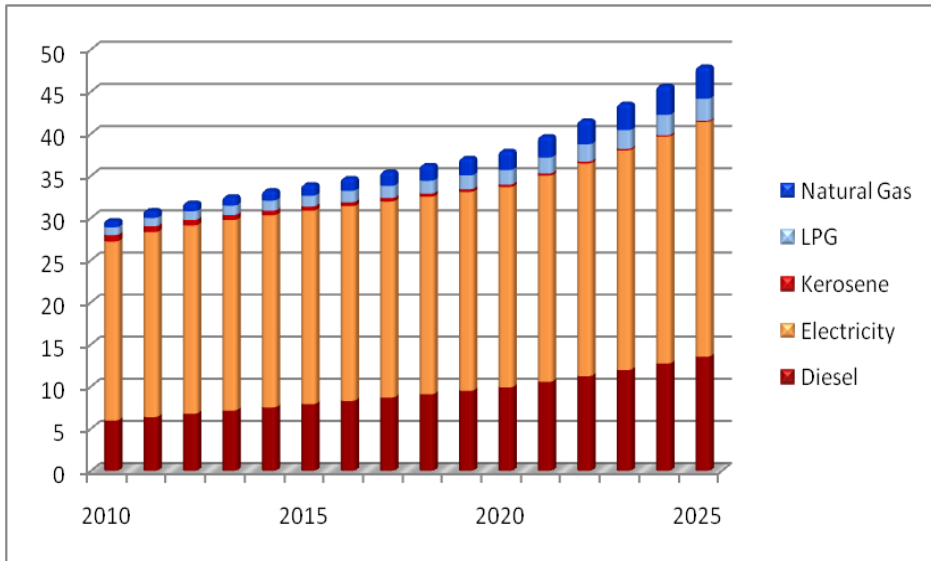
2. Energy Consumption in Residential Sector, 2011

(million BOE)



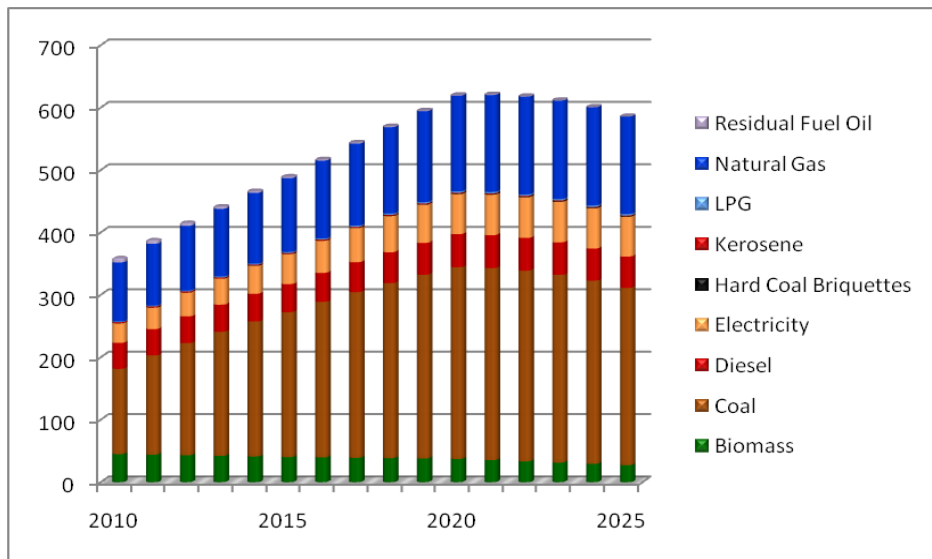
3. Energi Consumption in Commercial Sector, 2011

(million BOE)



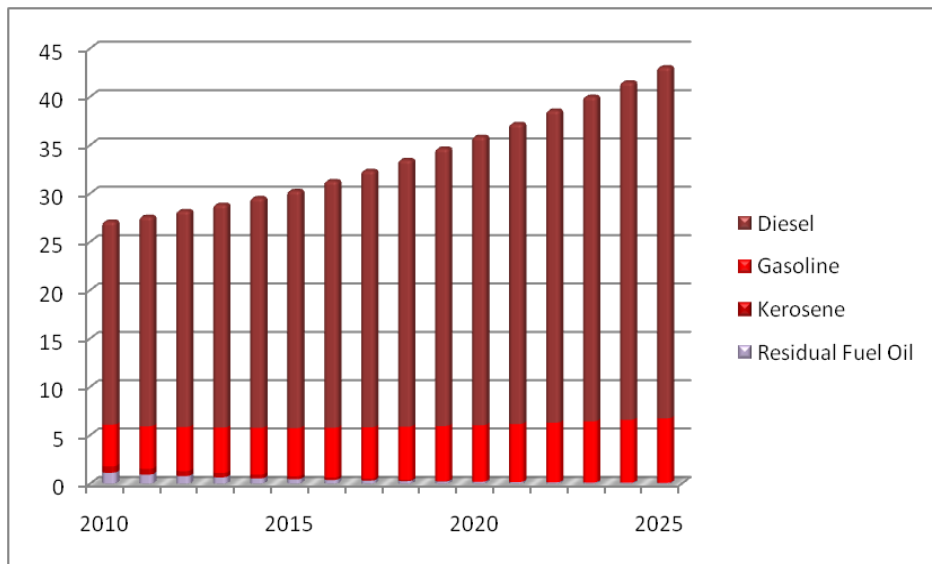
4. Energy Consumption in Industrial Sector, 2011

(Million BOE)



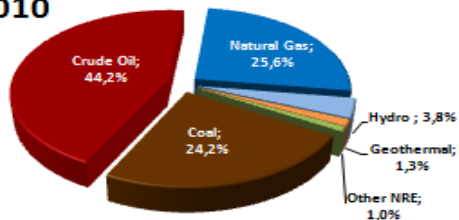
5. Energy Consumption in Other Sector, 2011

(million BOE)

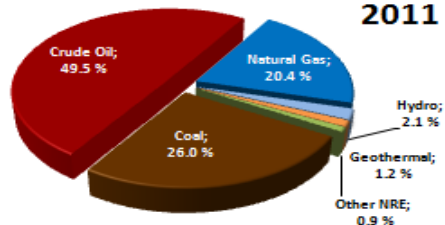


5. PRIMARY ENERGY MIX

2010

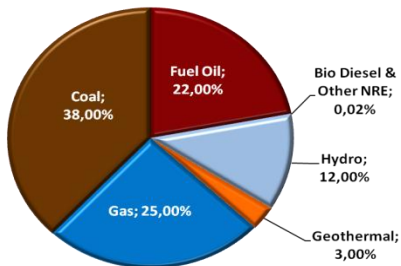


2011

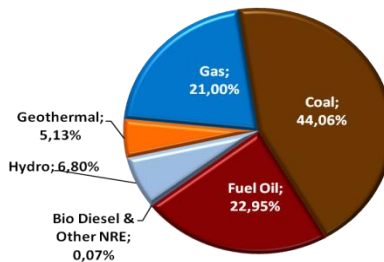


6. ENERGY MIX ON POWER SECTOR

2010



2011



ENERGY CONVERSION TABLE

Energy Type	Original Unit	Multiplier Factor to BOE	Multiplier Factor to TOE
Coal			
Anthracite	Ton	4,9893	0,6929
Imported Coal	Ton	4,2766	0,5939
Kalimantan Coal	Ton	4,2000	0,5833
Ombilin Coal	Ton	4,8452	0,6729
Tanjung Enim Coal	Ton	3,7778	0,5247
Lignite	Ton	3,0649	0,4257
Riau Peat	Ton	2,5452	0,3535
Briquette	Ton	3,5638	0,4949
Biomass			
Charcoal	Ton	4,9713	0,6904
Firewood	Ton	2,2979	0,3191
Natural Gas	MSCF	0,1796	0,0249
Gas Products			
City Gas	Thousand Kk	0,0007	0,0001
CNG	Thousand Kk	0,0007	0,0001
LNG	Ton	8,0532	1,1184
LNG	MMBTU	0,1796	0,0249
LPG	Ton	8,5246	1,1839
Oil			
Condensate	Barrel	0,9545	0,1326
Crude Oil	Barrel	1,0000	0,1389
Oil Fuel			
Aviation Gasoil (Avgas)	Kilo Litre	5,5530	0,7712
Aviation Turbin Gas (Avtur)	Kilo Litre	5,8907	0,8181
Super TT	Kilo Litre	5,8275	0,8093
Premix	Kilo Litre	5,8275	0,8093
Premium	Kilo Litre	5,8275	0,8093
Kerosene	Kilo Litre	5,9274	0,8232
ADO	Kilo Litre	6,4871	0,9009
IDO	Kilo Litre	6,6078	0,9177
FO	Kilo Litre	6,9612	0,9668
Oil Products			
Other Oil Products	Barrel	1,0200	0,1417
Refinery Fuel			
Refinery Fuel Gas (RFG)	Barrel	1,6728	0,2323
Refinery Fuel Oil (RFO)	Barrel	1,1236	0,1560
Feed Stock	Barrel	1,0423	0,1448
Geothermal	MWh	1,9558	0,2716
Hydro	MWh	2,5208	0,3501
Electric Power	MWh	0,6130	0,0851